

# SAI SANJAY BOMMISSETTY

Greater Chicago Area | (331) 385-5335 | ssb13@illinois.edu | linkedin.com/in/ssbommisetty

## EDUCATION

### University of Illinois Urbana-Champaign

B.S., Mechanical Sciences and Engineering | Minor, Hoefft Technology & Management

GPA: 3.86/4.00 | Dean's List

**Honors:** 1st Place (2026) & 2nd Place (2025), Engineering Industry Impact Award; 1st Place Distinguished Tech Award; Engineering Visionary Scholarship

**Relevant Coursework:** Mechanical Design I, Dynamics of Mechanical Systems, Engineering Materials, Design for Manufacturability, Introductory Solid Mechanics, Fundamentals of Fluid Dynamics, Thermodynamics, Introductory Dynamics, Electrical & Electronic Circuits, Computer-Aided Design

Aug 2024 – May 2028

Champaign, IL

## EXPERIENCE

### International Motors

Human Factors & Mechanical Design Intern

- Built a computer-vision gaze-tracking system achieving 2° accuracy using PCCR and stereo triangulation across synchronized multi-camera capture, mapping gaze onto a Unity-reconstructed 3D cab environment at 20 FPS
- Designed enclosures and hardware housings in Siemens NX with 6-DOF adjustable mounts and integrated cable routing; fabricated via 3D printing to meet simulator packaging constraints
- Analyzed human factors data from 30 participants across workshops and clinic studies, translating findings into thematic groupings and actionable design concepts
- Partnered with Experience Design and Clay Studio on material selection and layout planning in Figma, supporting concept sleeper-cab and interior mockups approved for downstream prototyping

May 2026 – Aug 2026

Lisle, IL

### Northrop Grumman Innovation Lab (NARB)

Research and Development Intern

- Led a 4-person team developing a wearable hardware system combining AR optics, custom PCB electronics, and sensor-fusion algorithms; owned mechanical design, rapid prototyping, and physical validation testing through to a functional prototype meeting accuracy and usability targets
- Designed and prototyped 3D CAD housings in SolidWorks integrating a Raspberry Pi, sensors, and bone-conduction transducers; verified electromechanical integration and sensor reliability across iterative build cycles, reducing redesign time

May 2025 – Oct 2025

Baltimore, MD

### LASSI – Laboratory for Advanced Space Systems at Illinois

Undergraduate Structures & Thermal Researcher

- Run FEA studies on the MonARCH 12U CubeSat deployer rail interface in Siemens NX, simulating structural loads and validating interface compliance against CubeSat standards
- Convert CAD geometry into Thermal Desktop models to evaluate temperature gradients and heat flux across satellite subsystems over orbital cycles, informing design changes under SWaP and radiation constraints

Sep 2025 – Present

Champaign, IL

## PROJECTS & LEADERSHIP

### ASME – Carle/CIMED Product Development Team

Team Lead

- Lead a 15-person team developing a smart mattress topper with embedded pressure-sensor arrays and pneumatic actuation to reduce pressure-ulcer risk in immobile patients; built data-collection hardware using Arduino, custom PCB design, and sensor conditioning
- Designed and tested a pneumatic repositioning system projected to reduce nurse repositioning workload by 30%; earned 1st and 2nd place in the Engineering Industry Impact Award among 100+ engineering projects

Sep 2024 – Apr 2026

Champaign, IL

### Open Source 4-Fingered Robotic Hand – UIUC Engineering Open House

Mechanical Team Member

- Contributed mechanical design and assembly to an open-source robotic hand for magnetic control of microbots, achieving 0.5 mm positional accuracy; integrated Arduino/ESP32 microcontrollers across the electromechanical system; team earned 1st Place Distinguished Tech Award among 100+ projects

Jan 2026 – Apr 2026

Champaign, IL

### International Consulting Network (ICON)

Project Manager & Membership Director

- Delivered consulting engagements for 4 international clients (PIMIC AI, fassforward, HTS Europe, MODO) spanning AI, hardware, and manufacturing; conducted market research and go-to-market analysis and presented findings to stakeholders across multiple countries
- Direct membership recruitment, onboarding, and expansion of the organization to additional universities

Jan 2025 – Present

Champaign, IL

## TECHNICAL SKILLS

**CAD & Simulation:** Siemens NX, SolidWorks, Fusion 360, AutoCAD, Ansys Thermal Desktop, Finite Element Analysis (FEA)

**Programming & Analysis:** Python, OpenCV, Arduino IDE

**Hardware & Prototyping:** PCB design, circuit design, sensor integration, embedded systems (ESP32 / Raspberry Pi), 3D printing, EmotiBit

**Design & Visualization:** Figma, ProtoPie, Unity, Blender, Microsoft Office

**Interests:** Humanoid robotics, space exploration, medical devices, physics, drawing, skiing, basketball, hiking